

Cultural Differences in Web Design on Chinese and Western Websites: A Literature Review

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Abstract – *Understanding cultural differences in web design is essential to facilitate cross-cultural communication and guide website localization. A literature review has been done with 17 studies that compare design elements of Chinese and Western websites. We found most results of the studies were difficult to compare, since different design elements or different attributes of the same elements were investigated. Only a few consistent findings could be drawn from the reviewed studies: (1) Chinese websites have more items and active elements, while Western websites have more footers; (2) Chinese websites use more red and blue colors, and links opening in new windows. By contrast, Western websites use more white color and underlined links; (3) Chinese websites have more images related to leaders or ceremonies, while Western websites have more images related to daily life or nature. We also found that most studies relate the differences to various dimensions of Hofstede's or Hall's cultural models, but the explanations sometimes raise questions. Based on the findings of this review, a discussion on future directions of cross-cultural design studies is presented.*

Index Terms - Chinese website, cultural differences, design elements, web design, Western website.

INTRODUCTION

People from different cultures have their specific preferences for web design because of distinct social norms and repetitive exposure [1]. Studies have confirmed that localized websites improve web usability and users' perception [2]–[4]. For instance, Faiola and Matei [2] found that American and Chinese users performed tasks faster while browsing websites created by designers from their own cultures. This finding leads to an interesting question: Which design elements are accustomed to people from a certain culture?

Previous research has demonstrated prevalence of different design elements by comparing websites from different countries. They label these country-specific

design features as “cultural markers”, which depict a country's cultural values manifested on websites. Moura *et al.* [1] conducted a literature review of the manifestation of cultural values and markers on websites from different countries. They found that, in most studies, Chinese and Western websites were compared, because of the distance between these two cultures [5,6]. Based on case studies, prior work has identified many different cultural markers prevalent on both sides. Due to fast-changing design fashions and web technology advances, it is valuable to review which findings remain consistent in the literature. However, a comprehensive understanding of cultural differences in web design on Chinese and Western websites is lacking.

Therefore, we conducted a literature review of studies that specifically compares design elements of Chinese and Western websites. The aim of this study is to investigate which cultural markers were identified in the literature and how they were explained from a cultural perspective.

BACKGROUND

There are two major approaches to explore which design elements are prevalent in a culture. The first one is the top-down approach, which develops analytical frameworks based on existing cultural models and compares design elements to the operationalized parameters of these frameworks. The other one is the bottom-up approach, which explores design characteristics first and then tries to connect the identified differences to more general cultural theories.

Most top-down studies derive their analytical frameworks from Hofstede's [7] or Hall's [8] cultural dimensions. They assume that the presence or high frequency of some design elements indicate certain cultural dimensions. For instance, guided navigation represents Hofstede's cultural dimension of uncertainty avoidance in Singh *et al.*'s framework, since users from high uncertainty avoidant cultures dislike ambiguous situations [9]. However, this top-down approach does not provide much insight into the seemingly numerous differences in web design, as most studies limit their focuses to a finite set of

web elements that represent a few cultural dimensions. This approach ignores other potential differences and excludes other possible cultural interpretations.

Compared to the top-down approach, the bottom-up approach can identify more nuanced differences in design elements and relate them to much broader cultural explanations. In a great number of bottom-up studies (see [1] for a literature review), the prevalence of different design elements in different countries have been identified and more cultural explanations have been applied. For instance, Alexander *et al.* [10] conducted a comparative study on cultural differences in web design among Australian, Chinese, and Saudi Arabian websites. They found that the design of Chinese websites was significantly different from that of Australian websites in many aspects. They connected these differences not only to Hofstede's and Hall's cultural models, but also to other cultural differences, such as different writing systems.

In conclusion, we believe that bottom-up studies can shed more light on cultural differences in web design on Chinese and Western websites. Therefore, we only selected studies that employed this approach and excluded the studies that used predefined analytical frameworks.

METHODOLOGY

I. Data collection

The current literature review includes 17 studies, within which design elements of Chinese and Western websites are compared. The reviewed studies were selected in a two-stage search process. First, we designed a broad search string as follows: Chin* AND cultur* AND web AND design. We then searched it in titles, abstracts, and keywords of articles in the databases of Scopus and Web of Science. The search was conducted in December 2021. The search results included 182 articles from Web of Science and 227 articles from Scopus (there was a considerable overlap between the results of both databases). Second, we scanned the articles manually and found many of them were irrelevant to our topic, due to our broad search string. We also noticed that some articles about usability or user experience tests on existing websites from both cultures. However, although they confirmed the influence of culture on users' perceptions, they did not specify which specific design elements affect participants' performance. So, they were not included in our literature review. In total, we selected 17 relevant studies ranging from 2003 to 2021.

II. Data analysis

Inspired by the literature review on website design by Moura *et al.* [1], each study was reviewed and synthesized according to the following categories: (1) author(s) and date; (2) aim of the study; (3) types of websites; (4) countries compared; (5) number of websites; (6) analytical method; (7) techniques to analyze the data; (8) cultural

models adopted to explain the differences; (9) categories of design elements; (10) different design elements identified; (11) main findings or differences found; and (12) cultural explanations of the findings. The main results are presented in the next section (contact the first author for the full results).

RESULTS

In this section, we first summarize the methodology of the reviewed studies to give an overview of the selected samples and research methods. Then, we describe the incomparable and inconsistent findings identified in these studies. Lastly, we illustrate the main consistent findings and their diverse cultural explanations.

I. Sample and methods

Sample Table 1 gives an overview of the selected samples of the articles [10]–[26]. In summary, nine studies can be classified as multinational comparative studies. Chinese websites were compared with these of a specific Western country in six studies. Among them, most ones compared websites between China and the U.S. One study compared Chinese and English versions of Chinese websites and another one classified websites by use of Chinese and English language. Regarding types of websites, most studies chose e-commerce or e-government websites, or a mix of these two categories. Two studies employed a different selection criterion: they selected websites by their popularity. Regarding sample size, most studies reported large-scale comparisons. Only in five studies, less than 50 samples were analyzed and in one study, the sample size was not mentioned.

TABLE 1. CHARACTERISTICS OF INCLUDED STUDIES.

Number of countries compared				
Single	Two	>Two	N/A	
1	6	9	1	
Types of websites				
Commerce	E-government	Mixed	Popularity	
9	4	2	2	
Sample size				
<10	10-50	51-100	>100	N/A
2	3	6	5	1

Methods All reviewed studies are case studies that analyzed the content of investigated websites, but did not include any information about the design guidelines. Frequencies or percentages of specific design elements were calculated and statistical tests were employed, such as chi-square tests or ANOVA tests. Most of them collected data manually. Only a few adopted automatic ways to collect data. For instance, [10] used the

combination of automatic and manual collection; and [16] used computational metrics to automatically quantify website colors.

II. Incomparable and inconsistent findings

Similar to [10]'s analytical framework, we categorized different design elements identified in reviewed studies into ten design categories, including layout, navigation, links, multimedia, color, visual representation, text formatting, language, privacy and content. We then compared the findings within each category. Most reviewed studies also adopted a similar categorization of design elements and could be fitted into one or several categories of our analytical framework.

However, we notice that most findings of the studies in each category are either incomparable or inconsistent. First, they are incomparable due to the measurements of different elements or different attributes of the same elements. For instance, several studies aimed to investigate different navigation styles between Chinese and Western websites. They took different design elements to represent navigation styles, such as menus, footers, sitemaps, or search bars. Furthermore, although several studies considered menus as an important indicator of the navigation style, they measured different attributes of menus: [10] measured the dynamics of the main menus; [11] counted the items in the main menus; [19] counted the

number of horizontal and vertical menus; [25] measured the number of multi-level menus. Measurements on different elements or diverse attributes of the same design elements make it difficult to compare their results. Second, we found many findings are inconsistent, even with the same attributes of the same design elements. For instance, the number of external links and internal links have been calculated in several studies. [10] identified that Chinese websites used more external links, while [18] found that English websites had more external links. By contrast, [15] and [23] showed no significant difference in external links among several countries. In terms of internal links, [18] and [21] found more internal links in Chinese websites, but [23] found no such a difference.

III. Consistent findings with diverse cultural explanations

Despite the incomparable and inconsistent findings, we have identified some consistent findings in six of the ten design categories (see Table 2). Most researchers related their findings to Hofstede's or Hall's cultural models. However, they employed different cultural dimensions of the two models to interpret the consistent findings and the interpretations were sometimes ambiguous. In this subsection, we demonstrate the main consistent findings and their diverse cultural interpretations.

TABLE 2. CONSISTENT DIFFERENCES IN WEB DESIGN ON CHINESE AND WESTERN WEBSITES.

Category	Chinese websites	Western websites	Cultural explanations	References
Layout	Denser with more items		Power distance; Long/short time orientation; Chinese ideograms	[10], [17], [22], [24]
	More (floating and clickable) banners		Uncertainty avoidance; High/low context;	[10],[12],[17], [23],[25]
Navigation		More footer navigation	Monochronic/polychronic time perception	[10],[23]
Links	More links opening in new windows		High/low context; Monochronic/polychronic time perception	[10], [13], [19]-[23]
	More font or color changed links	More underlined links	Chinese ideograms	[19]-[22], [27]
Multimedia	More pop-ups, videos, and animations (scrolling or rollover texts and images)		High/low context; Uncertainty avoidance;	[10], [12]- [14], [17], [19], [20], [25]
Color	More red and blue	More white	High/low context;	[10], [17],
	Low saturated colors	High saturated colors	Individualism/collectivism	[16],[24]
Visual representation	Images of leaders/families/ ceremonies/iconic/symbols/ accomplishments/monuments	Images of daily life/nature/ individuals	Individualism/collectivism; Masculinity/Femininity; Power distance; High/low context; Business culture	[10], [11], [13], [19], [20], [25]

Layout Several studies reported that Chinese websites are denser with more elements [10], [17], [22], [24] and contain more (clickable or floating) banners [10]. Regarding display density, [10] and [22] both found that Chinese websites had more items in the home page, including links, pictures, videos, titles, paragraphs, and captions. By dividing the total number of items by web page length, they also found that Chinese websites were denser. How to explain this phenomenon? [24] argued that Hofstede's cultural dimensions influenced website design in a complex and ambiguous way: with a long-term orientation, Chinese websites should have more information at the interface level. On the other hand, with a high-power distance, they should contain little information. Therefore, the author considered it was hard to decide why Chinese websites contained more elements. On the other hand, [22] concluded that Chinese writing has no separation between words and Chinese characters require less space, so more items can be stuffed into the interface. And [10] added that the high density of items provides more verbal and non-verbal cues for high-context cultures (such as Chinese culture) to communicate effectively. Another feature of Chinese websites was the heavy use of banners. Several studies identified more animated (clickable or floating) banners in Chinese websites [10], [12], [17], [23], [25]. [17] and [25] suggested that the heavy use of animated banners relates to the high uncertainty avoidance or high context in Chinese culture. These active elements could counteract uncertainty and bring entertaining, educational, and explorative experiences to Chinese users.

Navigation In this category, the studies mainly investigated the numbers or styles of menus, search bars and site maps. Only several studies identified that, on Chinese websites, footer navigation was not often present. Contrarily, on Western websites, footer navigation was frequently presented [10], [23]. According to the authors, the different use of footer navigation reflects the cultural difference between China's polychronic time perception and Western monochronic time perception [10], [23]. They argued that monochronic culture prefers to use fat footers and quicklinks as a way of linear navigation to direct users repeatedly and directly to the most important content.

Links The reviewed studies mainly explored the numbers, targets, forms, and styles of links. They have identified that Chinese websites had more color/text-changed links and more links opened in a new browser window, while Western websites had more underlined links [10], [13], [19]-[23], [27]. They argued that Chinese users, from a high-context and polychronic culture, prefer to navigate in parallel structures [10], [13], [19], [22]. According to them, more links opening in a new browser window could aid parallel browsing and allow users to switch among different pages. Next to this, a few studies examined the link click effects [19]-[22], [27]. They identified that Chinese websites preferred to change colors

or fonts of the links, while Western websites preferred to underline links. [22] stated that due to the visual nature of the Chinese ideograms, underlining links may hamper the readability of the links, so Chinese websites prefer to highlight links by color or font change.

Multimedia Multimedia means the integration of a variety of media, such as text, images, graphics, animation, audio, and video. The reviewed studies mainly focus on the presence and number of pop-ups, images, graphics, videos, animations, and sounds. They have demonstrated that Chinese websites used more pop-up windows, animations and videos [10], [12]-[14], [17], [19], [20], [25]. Besides, some studies looked at the animations of texts or images. They found more scrolling or rollover texts and images in Chinese websites [10], [17], [19], [20]. [17] and [25] claimed that the heavy use of animations could be attributed to Hofstede's cultural dimension of uncertainty avoidance. [10] and [13] explained that the heavy use of nontextual forms of communication provides context and aids Chinese users from the high-context culture to better understand a web page.

Color The numbers and saturation of foreground and background colors are often measured in the reviewed studies. They showed that Chinese websites used more red, blue, and low saturated colors, while Western websites used more white but high saturated colors [10], [16], [17], [24]. For instance, [10] calculated the ratios of the numbers of different colors occurred on the home page. They found Chinese websites had more blue and red colors, while Australian websites used more white. Regarding saturation of colors, [16] calculated the average pixel value for saturation among 44 countries. They found websites from the similar cultural clusters were similar in saturation. For eastern Asian countries, they had low average saturation of colors, while websites from Anglo-Saxon countries had the highest saturation. [27] proposed that high-context countries prefer to employ more of their traditional or national colors throughout websites. Red in traditional Chinese culture symbolizes happiness, good fortune and success, and blue stands for immortality and heaven, while white is the color of mourning [11], [17]. Regarding color saturation, [24] held that higher saturation to highlight information in US websites can be attributed to its high individualism. But it is difficult to see the relationship between the design elements of color and color saturation and the cultural dimensions of high/low context and individualism/collectivism.

Visual representation Visual representation refers to the styles and depictions of visuals. Reviewed studies have shown that images of leaders, families, ceremonies, iconic symbols, accomplishments, monuments were more common in Chinese websites. By contrast, images that portray daily life, nature and individuals were more frequent in Western websites [10], [11], [13], [19], [20], [25]. Several studies related the difference to the cultural dimensions of collectivism/individualism and high/low

power distance. They concluded that China's collectivistic culture has a high frequency of images presenting characteristics of collectivistic societies, such as families and traditions. China's high-power-distance culture also pays greater focus on leaders. On the other hand, the Western culture with low-power distance and individualism favors public images and individual life [10], [13], [20], [25]. By comparing differences between Chinese and American corporate websites, [11] argued that the phenomenon is due to different business cultures. Chinese are relationship-focused and they use images to boost their face, whereas Americans are deal-focused and they use images more directly related to their businesses.

DISCUSSION

The present review provides an up-to-date insight into the cultural differences in web design between Chinese and Western websites. Although only 17 studies have been examined, the review has identified different prevalence of design elements on Chinese and Western websites. The design features can be summarized as following: first, Chinese websites are more dynamic and explorative with more items and active elements. By contrast, Western websites are more textual and functional with more information in text and cluster of important links. Second, Chinese websites are more colorful with more red and blue colors, but less intense with more low saturated colors. On the other hand, Western websites are less colorful with more white, but more vivid with more highly saturated colors. Lastly, Chinese websites preferred to present images relating to collective or high-power-distance themes, while Western websites preferred to display images of individual or natural themes. The reoccurrences of the same findings in various studies reflect the most salient design features in Chinese and Western countries respectively. They can be seen as cultural markers that manifest different cultural dimensions in web design on Chinese and Western websites. The findings can contribute to a better understanding of cross-cultural web design.

However, we also notice that many findings of the reviewed studies are either inconsistent or incomparable. We think the main reason for the phenomenon is the lack of representative design elements in each design category and unclear descriptions of their measurements. These two problems are interrelated. Without clear descriptions of measurements of design elements, studies are hard to be replicated and future studies then may choose different design elements or attributes to measure. In this way, numerous design elements and their attributes might be selected and measured, which leads to the paucity of representative design elements and incongruent findings. For example, the design elements of both animation and banner were measured in several studies [10], [12], [17], [23], [25]. All of them treated animation and banner as separate elements in different categories to measure.

However, the attributes of banners can be mainly classified as static banners and animated banners. Studies did not specify how they count animated banners. So, some of them calculated the number of banners, while others counted the number of animated banners.

Future studies should choose representative (attributes of) design elements and give detailed descriptions of the measurements for future validation. Pauwels [28] suggested to collect and categorize an inventory of salient design element from previous studies and then choose representative design elements from the inventory. Concerning the attributes of the design elements, [28] also mentioned to classify the attributes of the design elements into fixed/static and moving/dynamic elements. The static attribute of a design element (for example, font size) focuses on the physical property of an element that influences the look of a web page. By contrast, the dynamic attribute of an element (for example, animated banners) involves users' interaction, which has an impact on the feel of a web page. These two aspects both influence users' perception on web design and potentially carry culturally specific meanings. Therefore, the attributes of design elements could be roughly classified and measured in the two aspects.

Future studies should also be aware of new design practices. The reviewed studies range from 2003 to 2021, which means many design elements may not exist or prevail at the time of investigation. Cross-cultural research on different practices of new design trends will be an interesting topic.

Another interesting finding in these studies is that Hofstede's or Hall's cultural models, functioning like panaceas, are used to explain a variety of situations, even conflicting ones. For instance, the heavy use of animation in Chinese websites has been identified in multiple studies. [17] claimed animation reduces uncertainty. It helps **counteract** the **high** uncertainty of Chinese users. On the other hand, [25] argued animation brings up unknown situations. Chinese users with **low** uncertainty avoidance **welcome** uncertain contexts. Following this logic, any finding can be interpreted as either counteract the effects of the cultural dimension or signify them. For them, the two cultural models seem to be flawless and can be applied in every design difference identified on the website. However, the revision of Hofstede's cultural model by himself poses doubts on the credibility of these arguments. Hofstede [29] originally ranked China as a high uncertainty avoidant culture and revised it to a low one in his latest theory in 2010 [7]. This is why in the last example, one study in 2005 said Chinese users avoids uncertainty, while another one in 2021 claimed that they welcome uncertainty. [28] also criticized the extended application of Hofstede's cultural model from the initial interpersonal context to extremely complex techno-cultural environments. It is doubtful to conclude numerous design differences into only a limited set of cultural dimensions.

Future studies might interpret the results by other cultural models, such as Nisbett's [30] or Lewis's [31] cultural models, since they mainly differentiate cultures not from the perspective of interpersonal communication, but from more explicit differences on behavior patterns and cognitive styles when dealing with surrounding environments. Or, they might come up with more plausible cultural interpretations depending on the context of their use in the new media. To achieve this goal, it is preferable to analyze the context of design process and compare design guidelines web designers follow. Cultural differences on design process and guidelines will be a fruitful area for future work.

We also believe it is imperative to validate and understand design differences from the users' perspective. When we collected the studies, we identified that many studies investigated the influence of different cultures on users' perceptions or performance on websites. For example, several studies show different cognitive styles between China and the West have an impact on users' viewing patterns and response on web pages [2], [3]. However, we excluded these studies, since they used existing websites that included a great number of variables which may affect users' perceptions. They are also not so informative about which design elements specifically impacts users' perceptions from different cultures. Only one reviewed study [22] conducted surveys and usability testing to validate the impact of design elements on users' performance and experience from different cultures. But it only used two existing sites to test five potential cultural markers out of many differences in design elements identified through the previous comparative study. It ignored many potential differences in existing websites that might influence users' perceptions and performance.

Future studies are recommended to use experimental sites to compare specific design elements and control for other variables. Through conducting usability or UX tests on these experimental sites and surveys on users from different cultures, cultural markers may be validated and their more implicit cultural meanings may be revealed.

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